

v1.0Photon Deep Dive Request

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: Luminous Monad Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0Photon Deep Dive Request V2.0.md

Revision Necessity — Four Drivers

- 1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
- 2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
- 3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^{17} + particles
- 4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Photon mass	Pre-2024 bound	Wu et al. Nature Astronomy 2024: $m_\gamma < 7.1 \times 10^{-23}$ eV from FRB lensing. 13× tighter	MEDIUM	Core photon propagator mass parameter updated
OAM modes	Helicity ± 1 only	V2.0: $L = 0, \pm 1, \dots, \pm 100$. Spiral worldlines for $L \neq 0$. NIST 100-qubit OAM 2023	HIGH	Missing 99 OAM modes is major omission for quantum information photon modeling
Superradiance	Not covered	Kerr BH superradiance: $E_{out}/E_{in} > 1$ from ergosphere reflection. New BH interaction type	LOW	New energy extraction mechanism previously unmodeled
BEC photons	Impossible in V1.0 (massless)	Bonn 2023: cavity photon BEC at room T — effective mass from confinement	MEDIUM	Thermalized photon condensate state requires new simulation mode

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** OAM
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged:** - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric ($C \cdot F \cdot S$ product formula) - Object-Oriented data structure (`struct ParticleInstance` extends to V2.0)

End of Versioning Delta | v1.0 Photon Deep Dive Request V1.0 archived | V2.0 is the active specification as of June 2026